FujiNet A500 integrated R3 — circuit review; NOT fabrication release

Protected USB input

3V3_AON
CC1
CC2
GND
VBUS_C
VBUS_PROTECTED

3V3_AON
CC1
CC2
GND
VBUS_C
VBUS_PROTECTED

File: input.kicad_sch

USB-C power and qualification

3V3
3V3_AON
3V3_PG
CC1
CC2
GND
VBUS_C
VBUS_PROTECTED

3V3
3V3_AON
3V3_PG
CC1
CC2
GND
VBUS_C
VBUS_PROTECTED

File: power.kicad_sch

USB routing and physical recovery

3V3
CC1
CC2
ESP_BOOT_N
ESP_DM
ESP_DP
ESP_EN
GND
RP_BOOT_N
RP_DM
RP_DP
RP_RUN
RUN_MODE
VBUS_C
VBUS_PROTECTED

3V3
CC1
CC2
ESP_BOOT_N
ESP_DM
ESP_DP
ESP_EN
GND
RP_BOOT_N
RP_DM
RP_DP
RP_RUN
RUN_MODE
VBUS_C
VBUS_PROTECTED

File: usb.kicad_sch

RP2350B and support

3V3
ACK_REQ
BUS_READY
DATA_DRIVE
GND
IRQ_REQ
RP_BOOT_N
RP_DM
RP_DP
RP_RUN
Z_A1
Z_A10
Z_A11
Z_A12
Z_A13
Z_A14
Z_A15
Z_A16
Z_A17
Z_A18
Z_A19
Z_A2
Z_A20
Z_A21
Z_A22
Z_A23
Z_A3
Z_A4
Z_A5
Z_A6
Z_A7
Z_A8
Z_A9
Z_AS_N
Z_CFGIN_N
Z_D0
Z_D1
Z_D10
Z_D11
Z_D12
Z_D13
Z_D14
Z_D15
Z_D2
Z_D3
Z_D4
Z_D5
Z_D6
Z_D7
Z_D8
Z_D9
Z_LDS_N
Z_READ
Z_UDS_N

3V3
ACK_REQ
BUS_READY
DATA_DRIVE
GND
IRQ_REQ
RP_BOOT_N
RP_DM
RP_DP
RP_RUN
Z_A1
Z_A10
Z_A11
Z_A12
Z_A13
Z_A14
Z_A15
Z_A16
Z_A17
Z_A18
Z_A19
Z_A2
Z_A20
Z_A21
Z_A22
Z_A23
Z_A3
Z_A4
Z_A5
Z_A6
Z_A7
Z_A8
Z_A9
Z_AS_N
Z_CFGIN_N
Z_D0
Z_D1
Z_D10
Z_D11
Z_D12
Z_D13
Z_D14
Z_D15
Z_D2
Z_D3
Z_D4
Z_D5
Z_D6
Z_D7
Z_D8
Z_D9
Z_LDS_N
Z_READ
Z_UDS_N

File: rp.kicad_sch

ESP32-S3 module and microSD

3V3
ESP_BOOT_N
ESP_DM
ESP_DP
ESP_EN
GND

3V3
ESP_BOOT_N
ESP_DM
ESP_DP
ESP_EN
GND

File: esp.kicad_sch

A500 socket and input buffers

3V3
AMIGA_5V
AMIGA_D0
AMIGA_D1
AMIGA_D10
AMIGA_D11
AMIGA_D12
AMIGA_D13
AMIGA_D14
AMIGA_D15
AMIGA_D2
AMIGA_D3
AMIGA_D4
AMIGA_D5
AMIGA_D6
AMIGA_D7
AMIGA_D8
AMIGA_D9
AMIGA_INT2_N
AMIGA_XRDY
GND
HOST_BERR_N
HOST_RESET_N
Z_A1
Z_A10
Z_A11
Z_A12
Z_A13
Z_A14
Z_A15
Z_A16
Z_A17
Z_A18
Z_A19
Z_A2
Z_A20
Z_A21
Z_A22
Z_A23
Z_A3
Z_A4
Z_A5
Z_A6
Z_A7
Z_A8
Z_A9
Z_AS_N
Z_CFGIN_N
Z_LDS_N
Z_READ
Z_UDS_N

3V3
AMIGA_5V
AMIGA_D0
AMIGA_D1
AMIGA_D10
AMIGA_D11
AMIGA_D12
AMIGA_D13
AMIGA_D14
AMIGA_D15
AMIGA_D2
AMIGA_D3
AMIGA_D4
AMIGA_D5
AMIGA_D6
AMIGA_D7
AMIGA_D8
AMIGA_D9
AMIGA_INT2_N
AMIGA_XRDY
GND
HOST_BERR_N
HOST_RESET_N
Z_A1
Z_A10
Z_A11
Z_A12
Z_A13
Z_A14
Z_A15
Z_A16
Z_A17
Z_A18
Z_A19
Z_A2
Z_A20
Z_A21
Z_A22
Z_A23
Z_A3
Z_A4
Z_A5
Z_A6
Z_A7
Z_A8
Z_A9
Z_AS_N
Z_CFGIN_N
Z_LDS_N
Z_READ
Z_UDS_N

File: address.kicad_sch

Independent byte-lane data buffers

3V3
AMIGA_D0
AMIGA_D1
AMIGA_D10
AMIGA_D11
AMIGA_D12
AMIGA_D13
AMIGA_D14
AMIGA_D15
AMIGA_D2
AMIGA_D3
AMIGA_D4
AMIGA_D5
AMIGA_D6
AMIGA_D7
AMIGA_D8
AMIGA_D9
GND
RX_HI_N
RX_LO_N
TX_HI_N
TX_LO_N
Z_D0
Z_D1
Z_D10
Z_D11
Z_D12
Z_D13
Z_D14
Z_D15
Z_D2
Z_D3
Z_D4
Z_D5
Z_D6
Z_D7
Z_D8
Z_D9

3V3
AMIGA_D0
AMIGA_D1
AMIGA_D10
AMIGA_D11
AMIGA_D12
AMIGA_D13
AMIGA_D14
AMIGA_D15
AMIGA_D2
AMIGA_D3
AMIGA_D4
AMIGA_D5
AMIGA_D6
AMIGA_D7
AMIGA_D8
AMIGA_D9
GND
RX_HI_N
RX_LO_N
TX_HI_N
TX_LO_N
Z_D0
Z_D1
Z_D10
Z_D11
Z_D12
Z_D13
Z_D14
Z_D15
Z_D2
Z_D3
Z_D4
Z_D5
Z_D6
Z_D7
Z_D8
Z_D9

File: data.kicad_sch

Hardware bus interlocks

3V3
3V3_PG
ACK_REQ
AMIGA_5V
AMIGA_INT2_N
AMIGA_XRDY
BUS_READY
DATA_DRIVE
GND
HOST_BERR_N
HOST_RESET_N
IRQ_REQ
RP_RUN
RUN_MODE
RX_HI_N
RX_LO_N
TX_HI_N
TX_LO_N
Z_A19
Z_A20
Z_A21
Z_A22
Z_A23
Z_AS_N
Z_LDS_N
Z_READ
Z_UDS_N

3V3
3V3_PG
ACK_REQ
AMIGA_5V
AMIGA_INT2_N
AMIGA_XRDY
BUS_READY
DATA_DRIVE
GND
HOST_BERR_N
HOST_RESET_N
IRQ_REQ
RP_RUN
RUN_MODE
RX_HI_N
RX_LO_N
TX_HI_N
TX_LO_N
Z_A19
Z_A20
Z_A21
Z_A22
Z_A23
Z_AS_N
Z_LDS_N
Z_READ
Z_UDS_N

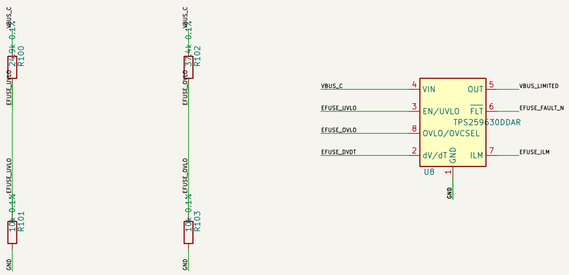
File: control.kicad_sch

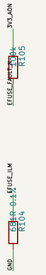
Sheet: /
File: integrated-a500-r3.kicad_sch

Title: FujiNet direct A500 integrated candidate

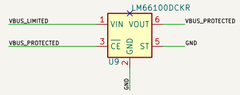
Size: A1	Date: 2026-11-09	**Rev: R3 candidate**
KiCad E.D.A. 10.0.7		Id: 1/9

USB VBUS -> current/voltage protection -> reverse-blocking ideal diode -> AON and qualified main supply.



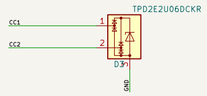












681R: nominal 1.337A limit. OVLO nominal 5.688V. Latch-off thermal fault: unplug USB to recover.

No Amiga supply connection. E-fuse pad/package and fast overvoltage transient review required before layout.



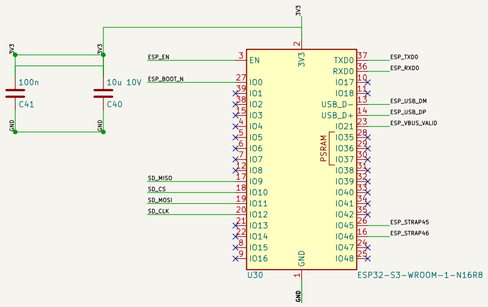
FujiNet NIO

Sheet: /Protected USB input/
File: input.kicad_sch

Title: Protected USB input

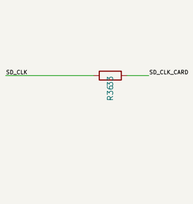
Size: A2	Date: 2026-11-09	**Rev: R3 candidate**
KiCad E.D.A. 10.0.7		Id: 2/9

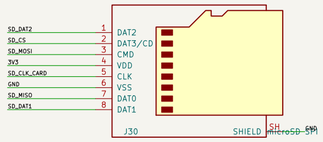
N16R8 module includes flash, PSRAM, crystal and RF. IO35/36/37 are reserved for octal PSRAM.

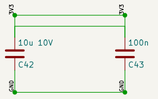


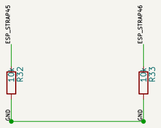


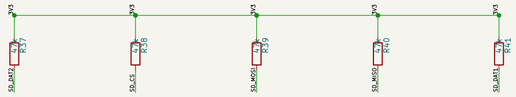














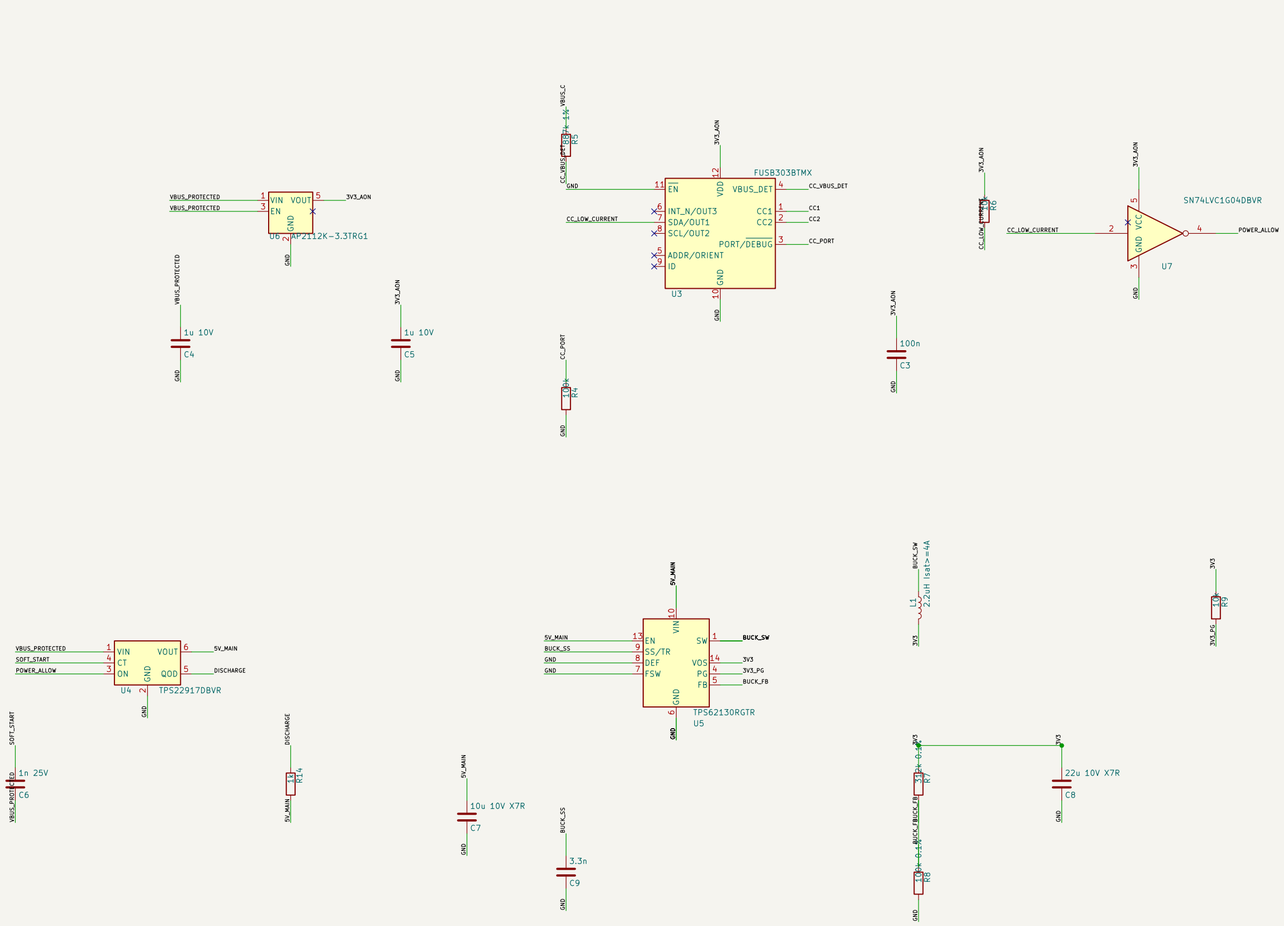
Antenna at enclosure/PCB edge: apply Espressif all-layer antenna keepout before PCB placement.

FujiNet NIO
Sheet: /ESP32-S3 module and microSD/
File: esp.kicad_sch

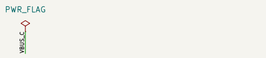
Title: ESP32-S3 module and microSD

Size: A2	Date: 2026-11-09	**Rev: R3 candidate**
KiCad E.D.A. 10.0.7		Id: 3/9

USB-C sink only. >=1.5 A advertised required. No Amiga supply connection to these rails.



POWER REVIEW OPEN: CC downgrade, input protection and thermal/capacitor derating. Not fabrication-ready.





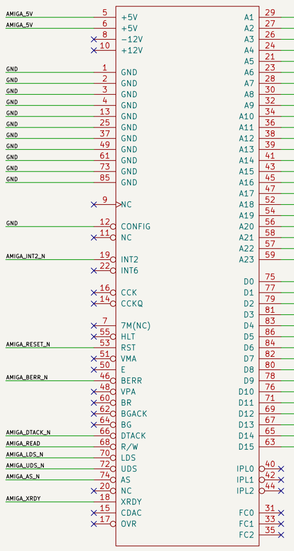


FujiNet NIO
Sheet: /USB-C power and qualification/
File: power.kicad_sch

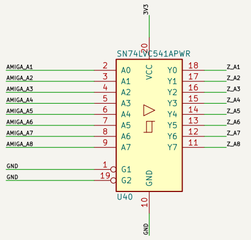
Title: USB-C power and qualification

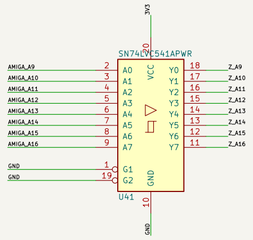
Size: A2	Date: 2026-11-09	Rev: R3 candidate
KiCad E.D.A. 10.0.7		Id: 4/9

86-contact A500 socket; NOT the existing male PCB-edge footprint. No A2000 SLAVE/CFGOUT here.



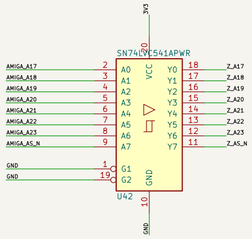
A500 FEMALE mating socket
J40

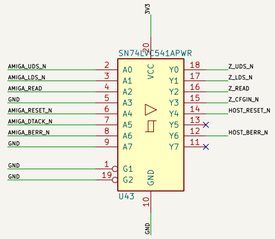
















U40–U43 use pin-compatible 541 logic symbol; fitted part MUST be 3.3V SN74LVC541A (5V tolerant, Ioff).

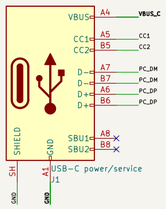
FujiNet NIO

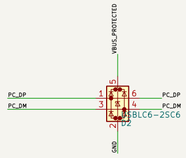
Sheet: /A500 socket and input buffers/
File: address.kicad_sch

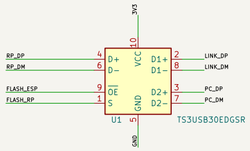
Title: A500 socket and input buffers

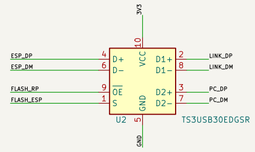
Size: A2	Date: 2026-11-09	**Rev: R3 candidate**
KiCad E.D.A. 10.0.7		Id: 5/9

Shared USB: cold mode changes only. Unplug power before moving the single MODE shunt.



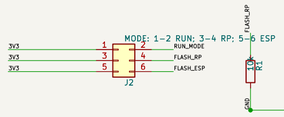


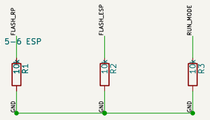






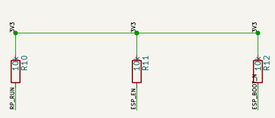


























Physical recovery: selected BOOT held low, peer held RESET. SW1/SW3 are independent resets.

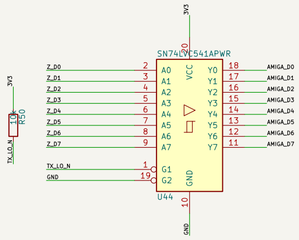
FujiNet NIO

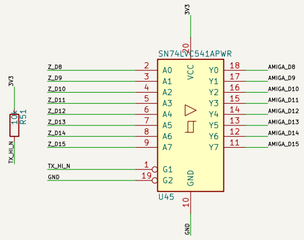
Sheet: /USB routing and physical recovery/
File: usb.kicad_sch

Title: USB routing and physical recovery

Size: A2	Date: 2026-11-09	**Rev: R3 candidate**
KiCad E.D.A. 10.0.7		Id: 6/9

Upper row: RP -> Amiga. Lower row: Amiga -> RP. Byte enables independent: hardware read/write qualification.

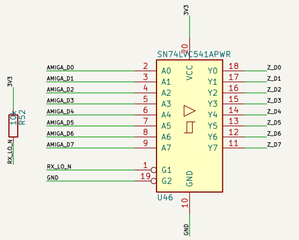


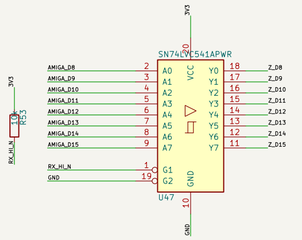
















PIO: preload read data/direction, pulse DATA_DRIVE, settle, pulse ACK_REQ. Return GPIO to input between cycles.

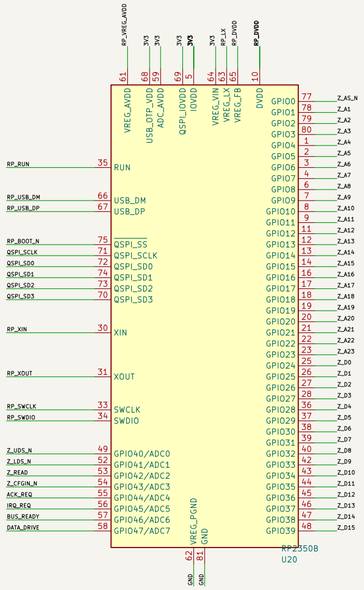
FujiNet NIO

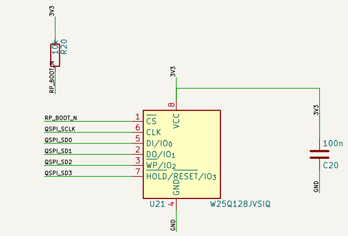
Sheet: /Independent byte-lane data buffers/
File: data.kicad_sch

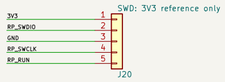
Title: Independent byte-lane data buffers

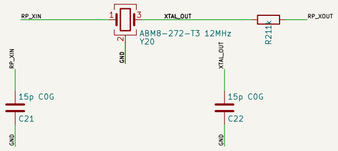
Size: A2	Date: 2026-11-09	**Rev: R3 candidate**
KiCad E.D.A. 10.0.7		Id: 7/9

48 GPIO: address/data/control only; USB and SWD dedicated pads. GPIO45 IRQ; GPIO46 BUS_READY.









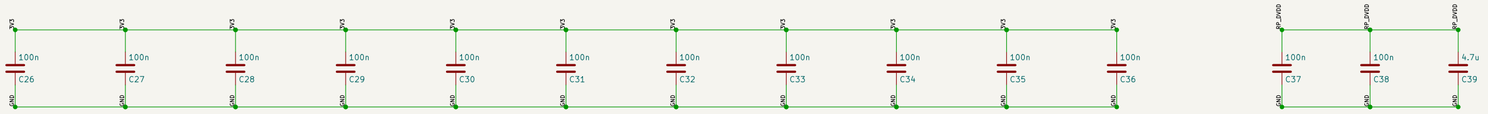




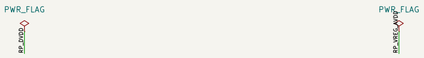




Core regulator placement MUST follow Raspberry Pi reference, including marked inductor orientation.



C26–C36: IOVDD/ADC/USB/QSPI decoupling. C37–C39: DVDD. Place at corresponding package pads.



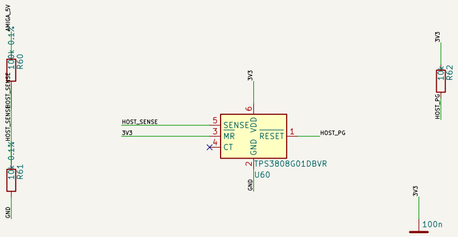
FujiNet NIO

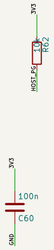
Sheet: /RP2350B and support/
File: rp.kicad_sch

Title: RP2350B and support

Size: A2	Date: 2026-11-09	**Rev: R3 candidate**
KiCad E.D.A. 10.0.7		Id: 8/9

Host monitor: 0.405V x 11 = 4.455V nominal trip; CT open = 20ms release delay. No host supply drawn for operation.



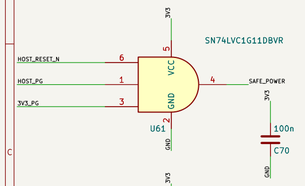


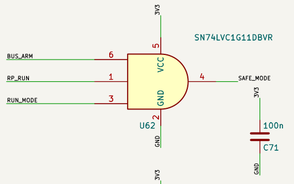
BUS ARM (normally OPEN)

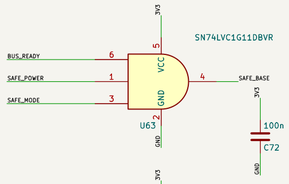


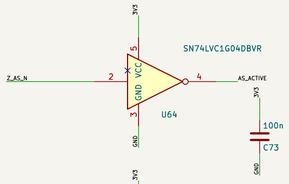


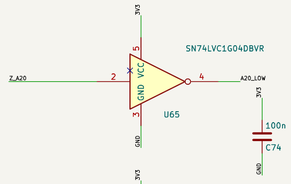


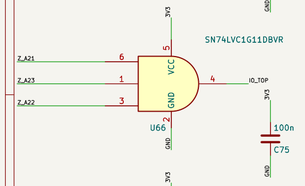


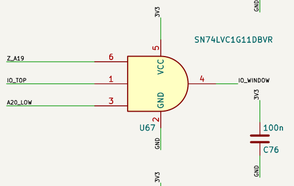


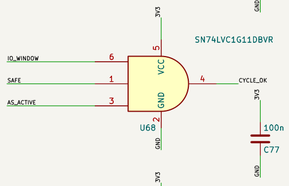


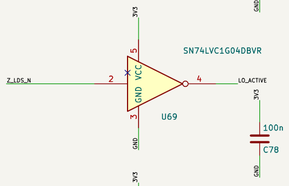


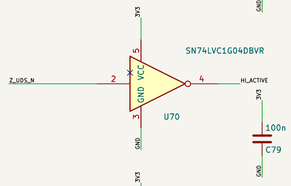


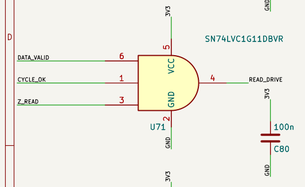


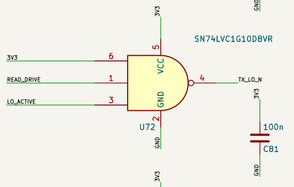


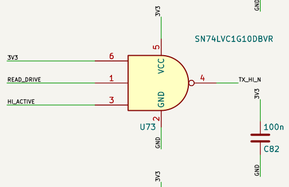


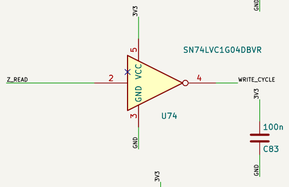


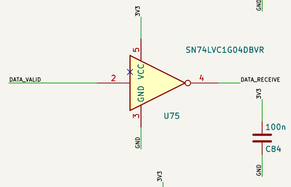


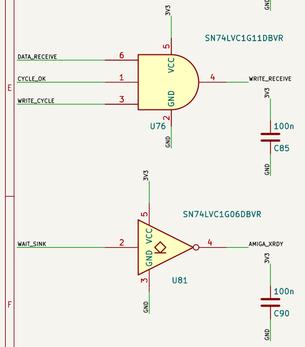


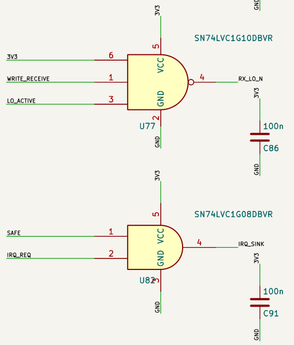


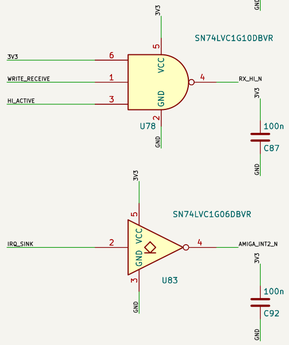


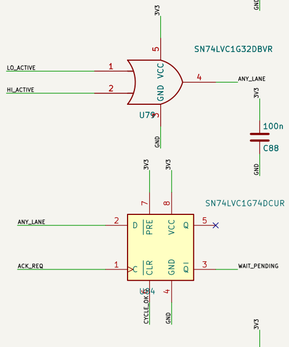


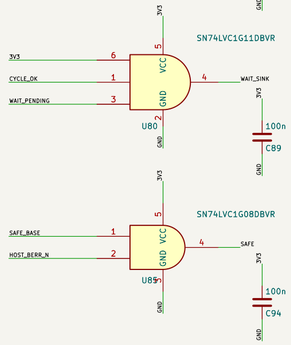


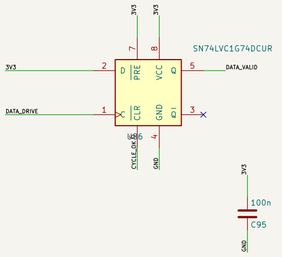












XRDY wait starts in hardware for E80000–EFFFFF; rising ACK_REQ releases this cycle. No local DTACK driver.

U84/U86 clear between cycles: stale ACK/DATA_DRIVE cannot carry into next cycle. BERR disables all outputs.

FujiNet NIO

Sheet: /Hardware bus interlocks/

File: control.kicad_sch

Title: Hardware bus interlocks

Size: A2	Date: 2026-11-09	**Rev: R3 candidate**
KiCad E.D.A. 10.0.7		Id: 9/9